A NONTRIVIAL POWER SUM OF DISTINCT FIELD ELEMENTS IS NONZERO AT SOME EXPONENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Together with the independence-embeddings lemma (b9 card L48386) this is the standard route from character independence to norm/trace nonvanishing statements.

Editorial note. The label lemma-sums-of-powers names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-independence-characters (source lines 1675-1713, monoid version): pairwise distinct monoid homomorphisms from a monoid G into a field L are L-linearly independent, i.e. for coefficients not all zero some g in G has a nonzero combination

Source reference. Original reference 0CKL: sources/stacks-fields/pinned-original-source.tex, source lines 1676–1676. Prerequisite pointer invoked at 1725; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L be a field. Let $n \geq 1$ and $\alpha_1, \ldots, \alpha_n \in L$ pairwise distinct elements of L. Then there exists an $e \geq 0$ such that $\sum_{i=1,\ldots,n} \alpha_i^e \neq 0$.*

Proof. Apply linear independence of characters (Lemma 0CKL) to the monoid homomorphisms $\mathbf{Z}_{\geq 0} \to L$, $e \mapsto \alpha_i^e$. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.